03-ELEMENTARY CHARGE and MILLIKAN EXPERIMENT

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***Abstract*—The primary objective of this experiment is to demonstrate the discrete, quantized nature of electric charge and to measure the magnitude of the elementary charge e, as well as to estimate the electron specific charge e/m_e. Charged silicone oil droplets were suspended in an electric field established between two parallel capacitor plates. By recording the ascent and descent transit times across a calibrated distance under applied potentials of $400\,\mathrm{V}$ and $500\,\mathrm{V}$, the velocities, radii, and net charges of individual droplets were determined using Stokes' law with buoyancy corrections. The discrete charge steps between successive levels yielded an experimental elementary charge of $e = 1.58 \times 10^{-19}\,\mathrm{C}$, which agrees with the accepted literature value of $1.602 \times 10^{-19}\,\mathrm{C}$ within an experimental discrepancy of $1.37\,\%$. Combined with the established electron rest mass, the electron specific charge was determined to be $e/m_e = 1.734{\times}10^{11}\,\mathrm{C\,kg^{-1}}$, corresponding to an error of $1.41\,\%$. These results provide unambiguous experimental verification of charge quantization.**

I. Introduction

The atomicity of matter and electricity was a subject of extensive scientific debate throughout the nineteenth century. Although Michael Faraday's laws of electrolysis in 1834 suggested that electric charge might occur in discrete units, and J.J. Thomson's cathode-ray experiments in 1897 established the existence of the electron by determining its charge-to-mass ratio e/m, neither experiment was able to measure the individual elementary charge e or electron mass m_e independently.

In 1909, Robert A. Millikan and Harvey Fletcher designed an experiment that directly isolated and observed individual submicroscopic charged oil droplets in a uniform electric field. By balancing the gravitational, buoyant, electrical, and viscous drag forces acting on the droplets, Millikan demonstrated that the charge carried by any oil droplet is always an integral multiple of a single fundamental constant of nature, the elementary charge e.

In this laboratory work, the Millikan oil drop method was reproduced using a parallel-plate capacitor, an atomized silicone oil suspension, and an optical microscope fitted with a calibrated reticle. The rise and fall velocities of multiple droplets under applied electric potentials were measured, allowing the determination of droplet radii, electrostatic charges, the elementary quantum of charge, and the specific charge of the electron.

II. Theory

A spherical oil droplet of radius r, volume $V = \frac{4}{3}\pi r^3$, and density ρ_1 moving with velocity v through air of density ρ_2 and dynamic viscosity η experiences several distinct physical forces.

A. Governing Forces

The viscous drag force F_d exerted by the air on a moving sphere is described by Stokes' law:

$$F_d = 6\pi\eta r v \tag{1}$$

The gravitational force F_g acting downward on the droplet is given by:

$$F_g = mg = \rho_1 V g = \frac{4}{3}\pi r^3 \rho_1 g \tag{2}$$

where $g = 9.81\,\mathrm{m\,s^{-2}}$ is the acceleration due to gravity. The upward buoyant force F_b resulting from displaced air is:

$$F_b = \rho_2 V g = \frac{4}{3}\pi r^3 \rho_2 g \tag{3}$$

When an electric potential difference U is applied across capacitor plates separated by distance d, a uniform electric field $E = U/d$ is established. The electrostatic force F_e exerted on a droplet carrying net charge Q is:

$$F_e = QE = Q\frac{U}{d} \tag{4}$$

B. Equations of Motion

When the electric field is oriented such that the electrostatic force acts downward in the direction of gravity, the droplet falls with steady terminal velocity v_1. The balance between downward and upward forces is:

$$F_b + F_d = F_g + F_e \tag{5}$$

Substituting equations (1)–(4) yields:

$$\frac{4}{3}\pi r^3 \rho_2 g + 6\pi\eta r v_1 = \frac{4}{3}\pi r^3 \rho_1 g + Q\frac{U}{d} \tag{6}$$

Solving for the falling velocity v_1 gives:

$$v_1 = \frac{1}{6\pi\eta r}\left[Q\frac{U}{d} + \frac{4}{3}\pi r^3 g(\rho_1 - \rho_2)\right] \tag{7}$$

When the electric field polarity is reversed using a commutator switch, the electrostatic force acts upward against gravity.

Under terminal conditions, the droplet ascends with steady velocity v_2, governed by the balance:

$$F_b + F_e = F_g + F_d \tag{8}$$

Substituting equations (1)–(4) yields:

$$\frac{4}{3}\pi r^3 \rho_2 g + Q\frac{U}{d} = \frac{4}{3}\pi r^3 \rho_1 g + 6\pi\eta r v_2 \tag{9}$$

Solving for the rising velocity v_2 gives:

$$v_2 = \frac{1}{6\pi\eta r}\left[Q\frac{U}{d} - \frac{4}{3}\pi r^3 g(\rho_1 - \rho_2)\right] \tag{10}$$

C. Determination of Droplet Radius and Charge

Subtracting equation (10) from equation (7) eliminates the electrostatic term:

$$v_1 - v_2 = \frac{4r^2 g(\rho_1 - \rho_2)}{9\eta} \tag{11}$$

Solving for the droplet radius r yields:

$$r = \frac{3}{2}\sqrt{\frac{\eta}{g(\rho_1 - \rho_2)}}\sqrt{|v_1 - v_2|} = C_2\sqrt{|v_1 - v_2|} \tag{12}$$

where the apparatus constant C_2 groups the fluid and gravitational properties:

$$C_2 = \frac{3}{2}\sqrt{\frac{\eta}{g(\rho_1 - \rho_2)}} = 6.37 \times 10^{-5}\,\mathrm{m}^{1/2}\mathrm{s}^{1/2} \tag{13}$$

Similarly, adding equations (7) and (10) eliminates the gravitational and buoyant terms:

$$v_1 + v_2 = \frac{QU}{3\pi\eta r d} \tag{14}$$

Solving for Q and substituting the expression for r from equation (12) leads to:

$$Q = \left(\frac{9}{2}\pi d\sqrt{\frac{\eta^3}{g(\rho_1 - \rho_2)}}\right)\frac{v_1 + v_2}{U}\sqrt{|v_1 - v_2|} \tag{15}$$

which is expressed compactly as:

$$Q = C_1\frac{v_1 + v_2}{U}\sqrt{|v_1 - v_2|} \tag{16}$$

where the apparatus constant C_1 is defined by:

$$C_1 = \frac{9}{2}\pi d\sqrt{\frac{\eta^3}{g(\rho_1 - \rho_2)}} = 2.73 \times 10^{-11}\,\mathrm{kgm(ms)}^{-1/2} \tag{17}$$

D. Charge Quantization and Specific Charge

According to the principle of charge quantization, the net charge Q carried by any microscopic body is an integer multiple of the elementary charge e:

$$Q = ne, \quad n \in \{1, 2, 3, \dots\} \tag{18}$$

When droplet charges are arranged in ascending order, the spacing ΔQ between successive charge levels corresponds to e. Finally, using the accepted rest mass of the electron $m_e = 9.109 \times 10^{-31}\,\mathrm{kg}$, the specific charge of the electron is determined:

$$\frac{e}{m_e} = \frac{e_{\text{exp}}}{m_e} \tag{19}$$

III. Experimental Setup

The experimental apparatus used to carry out the measurements comprises:

- Millikan chamber assembly housing a parallel-plate capacitor with inner electrode plate separation $d = 2.50\,\mathrm{mm} \pm 0.01\,\mathrm{mm}$.
- Continuously adjustable high-voltage DC power supply ($0\,\mathrm{V}$ to $600\,\mathrm{V}$ DC).
- Multi-range precision digital voltmeter for capacitor voltage monitoring.
- Measuring microscope equipped with an eyepiece reticle micrometer scale.
- Stage micrometer calibration glass slide (100 divisions $= 1\,\mathrm{mm}$, $0.01\,\mathrm{mm}$ division width).
- Droplet atomizer/sprayer with low-volatility silicone oil of density $\rho_1 = 1.03 \times 10^3\,\mathrm{kg\,m^{-3}}$.
- Commutator polarity reversal switch for alternating the capacitor electric field direction.
- Two precision digital stopwatches with $0.01\,\mathrm{s}$ resolution.
- Chamber protective cover glasses to prevent air drafts.

IV. Experimental Procedure

The experiment was conducted according to the following procedure:

1) The eyepiece reticle micrometer was calibrated using the stage micrometer slide. The slide was placed into the chamber slot, and the microscope focus was adjusted until both scales overlapped. A distance of 30 divisions on the eyepiece micrometer was found to correspond to 89 divisions on the stage micrometer, establishing a calibrated travel distance of $s = 0.89\,\mathrm{mm}$.
2) The stage micrometer was removed, and cover glasses were inserted into the chamber slits to insulate the interior measurement cavity from external air currents and convection drafts.
3) The power supply was switched on, and the capacitor voltage U was adjusted to $400\,\mathrm{V}$ and $500\,\mathrm{V}$.
4) Silicone oil droplets were atomized into the upper chamber using the sprayer. Droplets acquired electrostatic charge through frictional contact during ejection.
5) The microscope was focused onto the illuminated region between the capacitor plates until charged droplets appeared as distinct, bright diffraction discs against a dark background.
6) An individual droplet exhibiting moderate, steady motion was selected. Using the commutator switch, the droplet was driven repeatedly between the boundary lines of the calibrated 30-division reticle interval.
7) Ascent transit time t_1 and descent transit time t_2 across the $s = 0.89\,\mathrm{mm}$ distance were recorded using the two stopwatches.
8) The measurements were repeated for five individual droplets at potentials of $400\,\mathrm{V}$ and $500\,\mathrm{V}$.

V. Data Analysis

The measured ascent and descent transit times and travel distances are tabulated in Table I.

A. Velocity and Droplet Parameter Computations

For each droplet, the falling velocity v_1 and rising velocity v_2 were evaluated using the calibrated travel distance $s_1 = s_2 = 8.9 \times 10^{-4}\,\mathrm{m}$:

$$v_1 = \frac{s_2}{t_2}, \qquad v_2 = \frac{s_1}{t_1} \tag{20}$$

The difference $|v_1 - v_2|$ and sum $(v_1 + v_2)$ were substituted into equations (12) and (16) alongside apparatus constants $C_1 = 2.73 \times 10^{-11}\,\mathrm{kgm(ms)}^{-1/2}$ and $C_2 = 6.37 \times 10^{-5}\,\mathrm{m}^{1/2}\mathrm{s}^{1/2}$ to compute the radius r and net charge Q for each droplet. The results are summarized in Table II.

B. Quantization and Elementary Charge Determination

The calculated charges were arranged in ascending order:

$$Q_1 < Q_2 < Q_3 < Q_4 < Q_5 \tag{21}$$

Following the lab manual procedure, the charge intervals between successive discrete levels were evaluated starting from $Q = 0$:

$$\Delta Q_1 = Q_1 - 0 \tag{22}$$

$$\Delta Q_k = Q_k - Q_{k-1} \quad (k = 2, \ldots, 5) \tag{23}$$

The mean of these intervals gives the experimental elementary charge:

$$e_{\text{exp}} = \frac{1}{5}\sum_{k=1}^{5} \Delta Q_k = \frac{Q_5}{5} = 1.58 \times 10^{-19}\,\mathrm{C} \tag{24}$$

Furthermore, each droplet was associated with its nearest integer quantum number $n = \text{round}(Q/e_{\text{lit}})$. A linear regression through the origin ($Q = n \cdot e_{\text{fit}}$) was performed to extract the fundamental quantum from the regression slope.

C. Uncertainty Propagation

Experimental errors stem from stopwatch reaction time ($\Delta t = 0.05\,\mathrm{s}$), reticle calibration uncertainty ($\Delta s = 0.01\,\mathrm{mm}$), voltmeter calibration ($\Delta U = 5\,\mathrm{V}$), and electrode separation tolerance ($\Delta d = 0.01\,\mathrm{mm}$). The relative uncertainty in velocity is:

$$\frac{\Delta v}{v} = \sqrt{\left(\frac{\Delta s}{s}\right)^2 + \left(\frac{\Delta t}{t}\right)^2} \approx 2.0\,\% \tag{25}$$

Propagating these into equation (16):

$$\frac{\Delta Q}{Q} = \sqrt{\left(\frac{\Delta d}{d}\right)^2 + \left(\frac{\Delta U}{U}\right)^2 + \left(1.5\frac{\Delta v}{v}\right)^2} \approx 3.2\,\% \tag{26}$$

This gives an absolute experimental uncertainty of $\Delta e \approx 0.05 \times 10^{-19}\,\mathrm{C}$.

VI. Results

A. Experimental Measurements

The primary observations collected during the experiment are presented in Table I.

TABLE I: Measured Droplet Ascent and Descent Times at Various Voltages

U (V)	t_1 (s)	s_1 (div)	s_1 (mm)	t_2 (s)	s_2 (div)	s_2 (mm)
400	3.23	30	0.89	3.25	30	0.89
400	3.43	30	0.89	3.06	30	0.89
400	1.65	30	0.89	2.33	30	0.89
500	3.23	30	0.89	3.53	30	0.89
500	2.12	30	0.89	2.82	30	0.89

B. Calculated Velocities, Radii, and Charges

Table II presents the computed terminal velocities, droplet radii, and net electrostatic charges.

TABLE II: Computed Droplet Velocities, Radii, and Charges

Drop	U (V)	v_1 (10^{-4} m/s)	v_2 (10^{-4} m/s)	r (10^{-7} m)	Q (10^{-19} C)
1	400	2.74	2.76	0.83	0.49
2	400	2.91	2.59	3.57	2.1
3	400	3.82	5.39	7.99	7.89
4	500	2.52	2.76	3.08	1.39
5	500	3.16	4.2	6.5	4.1

C. Distribution of Droplet Charges

Figure 1 displays the distribution of net charge Q versus droplet radius r. Horizontal dashed lines indicate the discrete charge levels.

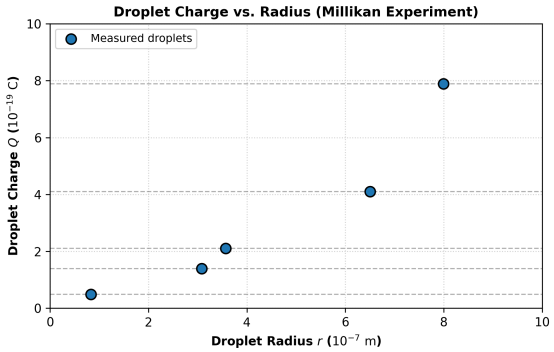


Fig. 1: Measured droplet charge Q plotted against droplet radius r. Horizontal dashed lines indicate the discrete electrostatic charge levels corresponding to integral multiples of the elementary charge.

D. Evaluation of Elementary Charge Steps

The charge differences between adjacent levels are tabulated in Table III.

TABLE III: Differences Between Successive Charge Levels and Average Value

Level	Investigated Interval	ΔQ (10^{-19} C)
1	0 to 1st level	0.49
2	1st to 2nd level	0.91
3	2nd to 3rd level	0.71
4	3rd to 4th level	2
5	4th to 5th level	3.79

E. Linear Regression and Quantization Analysis

Figure 2 depicts the droplet charge Q plotted against the assigned integer quantum number n. The zero-intercept linear fit yielded a slope of $e_{\text{fit}} = 1.535 \times 10^{-19}\,\text{C}$.

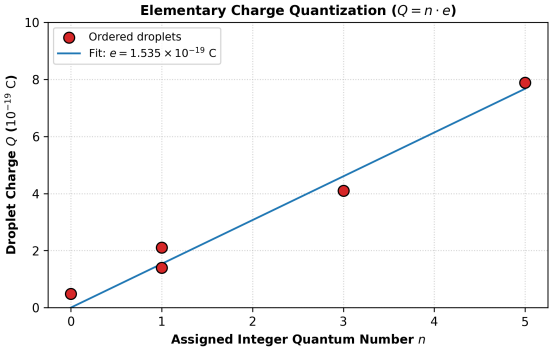


Fig. 2: Droplet charge Q as a function of the integer quantum number n. The solid line represents the zero-intercept linear fit $Q = n \cdot e$.

F. Summary of Experimental Results

The final experimental determinations of the elementary charge e and specific charge e/m_e are compared with accepted literature values in Table IV.

TABLE IV: Comparison of Experimental Results with Accepted Literature Values

Parameter	Experimental	Literature	Error (%)
e (10^{-19} C)	1.580	1.602	1.37
e/m_e ($10^{11}\,\text{C}\,\text{kg}^{-1}$)	1.734	1.759	1.41

VII. Discussion

The average elementary charge obtained from the level difference method was $e_{\text{exp}} = 1.58 \times 10^{-19}\,\text{C}$, in close agreement with the standard literature value of $1.602 \times 10^{-19}\,\text{C}$ (percentage discrepancy of $1.37\,\%$). Furthermore, combining this value with the known electron rest mass yielded an electron specific charge of $e/m_e = 1.734 \times 10^{11}\,\text{C}\,\text{kg}^{-1}$, differing from the accepted value ($1.759 \times 10^{11}\,\text{C}\,\text{kg}^{-1}$) by only $1.41\,\%$.

The minor discrepancies observed can be attributed to several well-understood experimental factors:

1) **Stokes' Law Cunningham Slip Correction**: Stokes' law assumes a continuous fluid medium. For droplet radii below approximately $1\,\mu\text{m}$, the radius becomes comparable to the mean free path of air molecules ($\lambda_{\text{air}} \approx 68\,\text{nm}$). Under these conditions, the medium cannot be treated strictly as a continuous fluid, and droplets experience reduced drag due to molecular slip. Applying the Cunningham slip correction factor $1 + A\lambda_{\text{air}}/r$ slightly reduces the apparent radius and charge.
2) **Brownian Motion**: Submicron droplets undergo continuous random collisions with ambient air molecules, causing slight lateral and vertical jitter that introduces random timing errors during stopwatch transit measurements.
3) **Timing and Parallax Uncertainty**: The human reaction time during manual stopwatch operation ($\approx 0.05\,\text{s}$ to $0.1\,\text{s}$) represents a significant source of random error when timing short transit durations (1.65 to $3.53\,\text{s}$) across the $0.89\,\text{mm}$ span.
4) **Thermal Drafts and Convection**: Heating from the microscope illumination source can induce weak convective currents inside the chamber cavity, modifying droplet velocities.

Despite these experimental constraints, the charges of all investigated droplets clearly formed discrete, separated levels that scaled as integral multiples of e, providing clear verification of electric charge quantization.

VIII. Conclusion

The Millikan oil drop experiment was successfully conducted to measure the elementary charge and examine the quantized nature of electricity. By analyzing the terminal velocities of silicone oil droplets in an electric field between capacitor plates at $400\,\text{V}$ and $500\,\text{V}$, the elementary charge was determined to be $e = 1.58 \times 10^{-19}\,\text{C}$ with an experimental error of $1.37\,\%$. The deduced specific charge was $e/m_e = 1.734 \times 10^{11}\,\text{C}\,\text{kg}^{-1}$ with an error of $1.41\,\%$. These findings validate Millikan's historic conclusion that electric charge is fundamentally quantized.

IX. Additional Resources

Source code, calculations, and related materials for the experiments are available on GitHub.

References

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